Dask for Machine Learning

Contents

Live Notebook

You can run this notebook in a live session launch binder or view it on Github.

This is a high-level overview demonstrating some the components of Dask-ML. Visit the main Dask-ML documentation, see the dask tutorial notebook 08, or explore some of the other machine-learning examples.

[1]:
```
from dask.distributed import Client, progress
client = Client(processes=False, threads_per_worker=4,
                n_workers=1, memory_limit='2GB')
client
```

[1]:

Client

Client-12a6896d-0de0-11ed-9ba6-000d3a8f7959

Connection method: Cluster object **Cluster type:** distributed.LocalCluster

Dashboard: http://10.1.1.64:8787/status

▸ Cluster Info

Distributed Training

Scikit-learn uses joblib for single-machine parallelism. This lets you train most estimators (anything that accepts an `n_jobs` parameter) using all the cores of your laptop or workstation.

Alternatively, Scikit-Learn can use Dask for parallelism. This lets you train those estimators using all the cores of your *cluster* without significantly changing your code.

This is most useful for training large models on medium-sized datasets. You may have a large model when searching over many hyper-parameters, or when using an ensemble method with many individual estimators. For too small datasets, training times will typically be small enough that cluster-wide parallelism isn't helpful. For too large datasets (larger than a single machine's memory), the scikit-learn estimators may not be able to cope (see below).

Create Scikit-Learn Estimator

[2]:
```
from sklearn.datasets import make_classification
from sklearn.svm import SVC
from sklearn.model_selection import GridSearchCV
import pandas as pd
```

We'll use scikit-learn to create a pair of small random arrays, one for the features `X`, and one for the target `y`.

[3]:
```
X, y = make_classification(n_samples=1000, random_state=0)
X[:5]
```

[3]:
```
array([[-1.06377997,  0.67640868,  1.06935647, -0.21758002,  0.46021477,
        -0.39916689, -0.07918751,  1.20938491, -0.78531472, -0.17218611,
        -1.08535744, -0.99311895,  0.30693511,  0.06405769, -1.0542328 ,
        -0.52749607, -0.0741832 , -0.35562842,  1.05721416, -0.90259159],
       [ 0.0708476 , -1.69528125,  2.44944917, -0.5304942 , -0.93296221,
         2.86520354,  2.43572851, -1.61850016,  1.30071691,  0.34840246,
         0.54493439,  0.22532411,  0.60556322, -0.19210097, -0.06802699,
         0.9716812 , -1.79204799,  0.01708348, -0.37566904, -0.62323644],
       [ 0.94028404, -0.49214582,  0.67795602, -0.22775445,  1.40175261,
         1.23165333, -0.77746425,  0.01561602,  1.33171299,  1.08477266,
        -0.97805157, -0.05012039,  0.94838552, -0.17342825, -0.47767184,
         0.76089649,  1.00115812, -0.06946407,  1.35904607, -1.18958963],
       [-0.29951677,  0.75988955,  0.18280267, -1.55023271,  0.33821802,
         0.36324148, -2.10052547, -0.4380675 , -0.16639343, -0.34083531,
         0.42435643,  1.17872434,  2.8314804 ,  0.14241375, -0.20281911,
         2.40571546,  0.31330473,  0.40435568, -0.28754632, -2.8478034 ],
       [-2.63062675,  0.23103376,  0.04246253,  0.47885055,  1.54674163,
         1.6379556 , -1.53207229, -0.73444479,  0.46585484,  0.4738362 ,
         0.98981401, -1.06119392, -0.88887952,  1.23840892, -0.57282854,
        -1.27533949,  1.0030065 , -0.47712843,  0.09853558,  0.52780407]])
```

We'll fit a Support Vector Classifier, using grid search to find the best value of the C hyperparameter.

[4]:
```
param_grid = {"C": [0.001, 0.01, 0.1, 0.5, 1.0, 2.0, 5.0, 10.0],
              "kernel": ['rbf', 'poly', 'sigmoid'],
              "shrinking": [True, False]}

grid_search = GridSearchCV(SVC(gamma='auto', random_state=0, probability=True),
                           param_grid=param_grid,
                           return_train_score=False,
                           cv=3,
                           n_jobs=-1)
```

To fit that normally, we would call

```
grid_search.fit(X, y)
```

To fit it using the cluster, we just need to use a context manager provided by joblib.

[5]:
```
import joblib

with joblib.parallel_backend('dask'):
    grid_search.fit(X, y)
```

We fit 48 different models, one for each hyper-parameter combination in `param_grid`, distributed across the cluster. At this point, we have a regular scikit-learn model, which can be used for prediction, scoring, etc.

[6]:
```
pd.DataFrame(grid_search.cv_results_).head()
```

			Dask	Distributed		Dask ML	Examples		Ecosystem	Comm
0	0.267177	0.011928	0.030591	0.006284	0.001	rbf	True	'rbf', 'shrinking': True}	0.502994	0.501
1	0.263738	0.005183	0.028903	0.005411	0.001	rbf	False	{'C': 0.001, 'kernel': 'rbf', 'shrinking': False}	0.502994	0.501
2	0.193400	0.005858	0.018773	0.003098	0.001	poly	True	{'C': 0.001, 'kernel': 'poly', 'shrinking': True}	0.502994	0.501
3	0.196229	0.006628	0.018826	0.002250	0.001	poly	False	{'C': 0.001, 'kernel': 'poly', 'shrinking': Fa...	0.502994	0.501
4	0.378106	0.006729	0.037400	0.002586	0.001	sigmoid	True	{'C': 0.001, 'kernel': 'sigmoid', 'shrinking':...	0.502994	0.501

[7]:
```
grid_search.predict(X)[:5]
```

[7]:
```
array([0, 1, 1, 1, 0])
```

[8]:
```
grid_search.score(X, y)
```

[8]:
```
0.983
```

For more on training scikit-learn models with distributed joblib, see the dask-ml documentation.

Training on Large Datasets

Most estimators in scikit-learn are designed to work on in-memory arrays. Training with larger datasets may require different algorithms.

All of the algorithms implemented in Dask-ML work well on larger than memory datasets, which you might store in a dask array or dataframe.

[9]:
```
%matplotlib inline
```

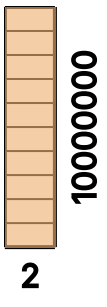
[10]:
```
import dask_ml.datasets
import dask_ml.cluster
import matplotlib.pyplot as plt
```

In this example, we'll use `dask_ml.datasets.make_blobs` to generate some random *dask* arrays.

[11]:
```
X, y = dask_ml.datasets.make_blobs(n_samples=10000000,
                                   chunks=1000000,
                                   random_state=0,
                                   centers=3)
X = X.persist()
X
```

[11]:

	Array	Chunk
Bytes	152.59 MiB	15.26 MiB
Shape	(10000000, 2)	(1000000, 2)
Count	10 Tasks	10 Chunks
Type	float64	numpy.ndarray



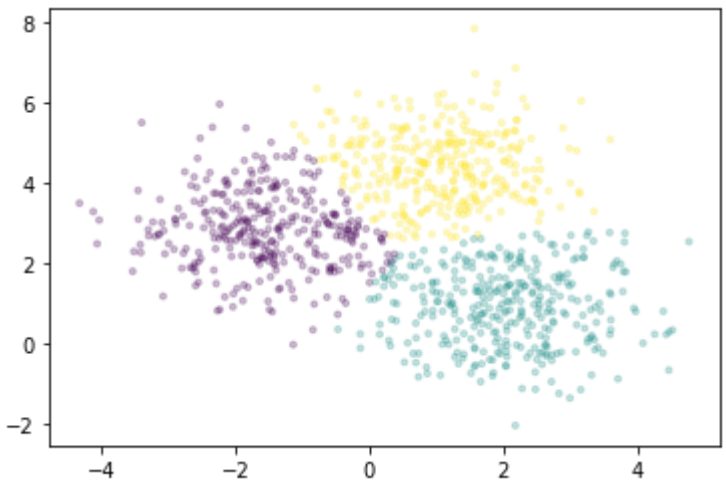
We'll use the k-means implemented in Dask-ML to cluster the points. It uses the `k-means||` (read: "k-means parallel") initialization algorithm, which scales better than `k-means++`. All of the computation, both during and after initialization, can be done in parallel.

```
/usr/share/miniconda3/envs/dask-examples/lib/python3.9/site-
packages/dask/base.py:1283: UserWarning: Running on a single-machine
scheduler when a distributed client is active might lead to unexpected
results.
  warnings.warn(
```

[12]:
```
KMeans(init_max_iter=2, n_clusters=3, oversampling_factor=10)
```

We'll plot a sample of points, colored by the cluster each falls into.

[13]:
```
fig, ax = plt.subplots()
ax.scatter(X[::10000, 0], X[::10000, 1], marker='.', c=km.labels_[::10000],
           cmap='viridis', alpha=0.25);
```



For all the estimators implemented in Dask-ML, see the API documentation.

By Dask Developers